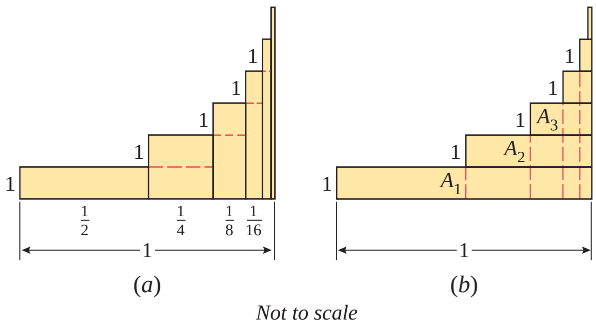
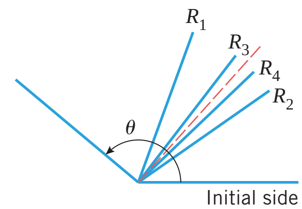
part (b) the configuration in part (a) has been divided into rectangles with areas $A_1, A_2, A_3, \ldots$. Find the sum $A_1 + A_2 + A_3 + \cdots$.



▲ **Figure Ex-38**

39. As shown in the accompanying figure, suppose that an angle θ is bisected using a straightedge and compass to produce ray R_1, then the angle between R_1 and the initial side is bisected to produce ray R_2. Thereafter, rays $R_3, R_4, R_5, \ldots$ are constructed in succession by bisecting the angle between the preceding two rays. Show that the sequence of angles that these rays make with the initial side has a limit of $\theta/3$.

Source: This problem is based on "Trisection of an Angle in an Infinite Number of Steps" by Eric Kincannon, which appeared in *The College Mathematics Journal*, Vol. 21, No. 5, November 1990.



◀ **Figure Ex-39**

C **40.** In each part, use a CAS to find the sum of the series if it converges, and then confirm the result by hand calculation.

(a) $\displaystyle\sum_{k=1}^{\infty}(-1)^{k+1}2^k 3^{2-k}$ (b) $\displaystyle\sum_{k=1}^{\infty}\frac{3^{3k}}{5^{k-1}}$ (c) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{4k^2-1}$

41. Writing Discuss the similarities and differences between what it means for a sequence to converge and what it means for a series to converge.

42. Writing Read about Zeno's dichotomy paradox in an appropriate reference work and relate the paradox in a setting that is familiar to you. Discuss a connection between the paradox and geometric series.

✔ QUICK CHECK ANSWERS 9.3

1. sequence; series **2.** $\frac{1}{2}; \frac{3}{4}; \frac{7}{8}; \frac{15}{16}; 1 - \frac{1}{2^n}$ **3.** The sequence of partial sums converges.
4. ar^k $(a \neq 0)$; $\frac{a}{1-r}$; $|r| < 1$; $|r| \geq 1$ **5.** $\frac{1}{k}$; diverge

9.4 CONVERGENCE TESTS

In the last section we showed how to find the sum of a series by finding a closed form for the nth partial sum and taking its limit. However, it is relatively rare that one can find a closed form for the nth partial sum of a series, so alternative methods are needed for finding the sum of a series. One possibility is to prove that the series converges, and then to approximate the sum by a partial sum with sufficiently many terms to achieve the desired degree of accuracy. In this section we will develop various tests that can be used to determine whether a given series converges or diverges.

THE DIVERGENCE TEST

In stating general results about convergence or divergence of series, it is convenient to use the notation $\sum u_k$ as a generic notation for a series, thus avoiding the issue of whether the sum begins with $k = 0$ or $k = 1$ or some other value. Indeed, we will see shortly that the starting index value is irrelevant to the issue of convergence. The kth term in an infinite series $\sum u_k$ is called the ***general term*** of the series. The following theorem establishes a relationship between the limit of the general term and the convergence properties of a series.

> **9.4.1 THEOREM** (*The Divergence Test*)
>
> (*a*) *If* $\lim_{k\to+\infty} u_k \neq 0$, *then the series* $\sum u_k$ *diverges.*
>
> (*b*) *If* $\lim_{k\to+\infty} u_k = 0$, *then the series* $\sum u_k$ *may either converge or diverge.*

PROOF (*a*) To prove this result, it suffices to show that if the series converges, then $\lim_{k\to+\infty} u_k = 0$ (why?). We will prove this alternative form of (*a*).

Let us assume that the series converges. The general term u_k can be written as

$$u_k = s_k - s_{k-1} \tag{1}$$

where s_k is the sum of the terms through u_k and s_{k-1} is the sum of the terms through u_{k-1}. If S denotes the sum of the series, then $\lim_{k\to+\infty} s_k = S$, and since $(k-1)\to+\infty$ as $k\to+\infty$, we also have $\lim_{k\to+\infty} s_{k-1} = S$. Thus, from (1)

$$\lim_{k\to+\infty} u_k = \lim_{k\to+\infty} (s_k - s_{k-1}) = S - S = 0$$

PROOF (*b*) To prove this result, it suffices to produce both a convergent series and a divergent series for which $\lim_{k\to+\infty} u_k = 0$. The following series both have this property:

$$\frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^k} + \cdots \quad \text{and} \quad 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} + \cdots$$

The first is a convergent geometric series and the second is the divergent harmonic series. ■

WARNING

The converse of Theorem 9.4.2 is false; that is, showing that

$$\lim_{k\to+\infty} u_k = 0$$

does not prove that $\sum u_k$ converges, since this property may hold for divergent as well as convergent series. This is illustrated in the proof of part (*b*) of Theorem 9.4.1.

The alternative form of part (*a*) given in the preceding proof is sufficiently important that we state it separately for future reference.

> **9.4.2 THEOREM** *If the series* $\sum u_k$ *converges, then* $\lim_{k\to+\infty} u_k = 0$.

▶ Example 1 The series

$$\sum_{k=1}^{\infty} \frac{k}{k+1} = \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \cdots + \frac{k}{k+1} + \cdots$$

diverges since

$$\lim_{k\to+\infty} \frac{k}{k+1} = \lim_{k\to+\infty} \frac{1}{1+1/k} = 1 \neq 0 \blacktriangleleft$$

■ ALGEBRAIC PROPERTIES OF INFINITE SERIES

For brevity, the proof of the following result is omitted.

See Exercises 27 and 28 for an exploration of what happens when $\sum u_k$ or $\sum v_k$ diverge.

9.4.3 THEOREM

(*a*) *If* $\sum u_k$ *and* $\sum v_k$ *are convergent series, then* $\sum(u_k + v_k)$ *and* $\sum(u_k - v_k)$ *are convergent series and the sums of these series are related by*

$$\sum_{k=1}^{\infty}(u_k + v_k) = \sum_{k=1}^{\infty} u_k + \sum_{k=1}^{\infty} v_k$$
$$\sum_{k=1}^{\infty}(u_k - v_k) = \sum_{k=1}^{\infty} u_k - \sum_{k=1}^{\infty} v_k$$

(*b*) *If c is a nonzero constant, then the series* $\sum u_k$ *and* $\sum cu_k$ *both converge or both diverge. In the case of convergence, the sums are related by*

$$\sum_{k=1}^{\infty} cu_k = c\sum_{k=1}^{\infty} u_k$$

(*c*) *Convergence or divergence is unaffected by deleting a finite number of terms from a series; in particular, for any positive integer* K, *the series*

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \cdots$$
$$\sum_{k=K}^{\infty} u_k = u_K + u_{K+1} + u_{K+2} + \cdots$$

both converge or both diverge.

WARNING

Do not read too much into part (*c*) of Theorem 9.4.3. Although convergence is not affected when finitely many terms are deleted from the beginning of a convergent series, the *sum* of the series is changed by the removal of those terms.

Example 2 Find the sum of the series

$$\sum_{k=1}^{\infty}\left(\frac{3}{4^k} - \frac{2}{5^{k-1}}\right)$$

Solution. The series

$$\sum_{k=1}^{\infty}\frac{3}{4^k} = \frac{3}{4} + \frac{3}{4^2} + \frac{3}{4^3} + \cdots$$

is a convergent geometric series $\left(a = \frac{3}{4}, r = \frac{1}{4}\right)$, and the series

$$\sum_{k=1}^{\infty}\frac{2}{5^{k-1}} = 2 + \frac{2}{5} + \frac{2}{5^2} + \frac{2}{5^3} + \cdots$$

is also a convergent geometric series $\left(a = 2, r = \frac{1}{5}\right)$. Thus, from Theorems 9.4.3(*a*) and 9.3.3 the given series converges and

$$\sum_{k=1}^{\infty}\left(\frac{3}{4^k} - \frac{2}{5^{k-1}}\right) = \sum_{k=1}^{\infty}\frac{3}{4^k} - \sum_{k=1}^{\infty}\frac{2}{5^{k-1}}$$
$$= \frac{\frac{3}{4}}{1-\frac{1}{4}} - \frac{2}{1-\frac{1}{5}} = -\frac{3}{2} \blacktriangleleft$$

Example 3 Determine whether the following series converge or diverge.

$$\text{(a)} \sum_{k=1}^{\infty} \frac{5}{k} = 5 + \frac{5}{2} + \frac{5}{3} + \cdots + \frac{5}{k} + \cdots \qquad \text{(b)} \sum_{k=10}^{\infty} \frac{1}{k} = \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \cdots$$

Solution. The first series is a constant times the divergent harmonic series, and hence diverges by part (*b*) of Theorem 9.4.3. The second series results by deleting the first nine terms from the divergent harmonic series, and hence diverges by part (*c*) of Theorem 9.4.3. ◀

THE INTEGRAL TEST

The expressions

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \quad \text{and} \quad \int_1^{+\infty} \frac{1}{x^2}\, dx$$

are related in that the integrand in the improper integral results when the index k in the general term of the series is replaced by x and the limits of summation in the series are replaced by the corresponding limits of integration. The following theorem shows that there is a relationship between the convergence of the series and the integral.

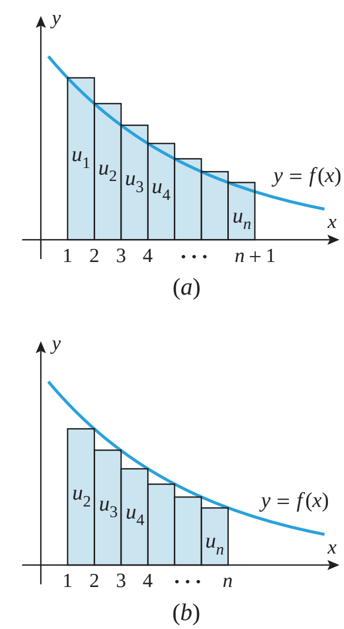


Figure 9.4.1

> **9.4.4 THEOREM (*The Integral Test*)** *Let $\sum u_k$ be a series with positive terms. If f is a function that is decreasing and continuous on an interval $[a, +\infty)$ and such that $u_k = f(k)$ for all $k \geq a$, then*
>
> $$\sum_{k=1}^{\infty} u_k \quad \textit{and} \quad \int_a^{+\infty} f(x)\, dx$$
>
> *both converge or both diverge.*

The proof of the integral test is deferred to the end of this section. However, the gist of the proof is captured in Figure 9.4.1: if the integral diverges, then so does the series (Figure 9.4.1*a*), and if the integral converges, then so does the series (Figure 9.4.1*b*).

Example 4 Show that the integral test applies, and use the integral test to determine whether the following series converge or diverge.

$$\text{(a)} \sum_{k=1}^{\infty} \frac{1}{k} \qquad \text{(b)} \sum_{k=1}^{\infty} \frac{1}{k^2}$$

Solution (a). We already know that this is the divergent harmonic series, so the integral test will simply illustrate another way of establishing the divergence.

Note first that the series has positive terms, so the integral test is applicable. If we replace k by x in the general term $1/k$, we obtain the function $f(x) = 1/x$, which is decreasing and continuous for $x \geq 1$ (as required to apply the integral test with $a = 1$). Since

$$\int_1^{+\infty} \frac{1}{x}\, dx = \lim_{b \to +\infty} \int_1^b \frac{1}{x}\, dx = \lim_{b \to +\infty} [\ln b - \ln 1] = +\infty$$

the integral diverges and consequently so does the series.

WARNING

In part (b) of Example 4, do not erroneously conclude that the sum of the series is 1 because the value of the corresponding integral is 1. You can see that this is not so since the sum of the first two terms alone exceeds 1. Later, we will see that the sum of the series is actually $\pi^2/6$.

Solution (b). Note first that the series has positive terms, so the integral test is applicable. If we replace k by x in the general term $1/k^2$, we obtain the function $f(x) = 1/x^2$, which is decreasing and continuous for $x \geq 1$. Since

$$\int_1^{+\infty} \frac{1}{x^2}\,dx = \lim_{b\to+\infty}\int_1^b \frac{dx}{x^2} = \lim_{b\to+\infty}\left[-\frac{1}{x}\right]_1^b = \lim_{b\to+\infty}\left[1-\frac{1}{b}\right] = 1$$

the integral converges and consequently the series converges by the integral test with $a = 1$. ◀

■ *p*-SERIES

The series in Example 4 are special cases of a class of series called ***p-series*** or ***hyperharmonic series***. A p-series is an infinite series of the form

$$\sum_{k=1}^{\infty}\frac{1}{k^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{k^p} + \cdots$$

where $p > 0$. Examples of p-series are

$$\sum_{k=1}^{\infty}\frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} + \cdots \qquad \boxed{p = 1}$$

$$\sum_{k=1}^{\infty}\frac{1}{k^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{k^2} + \cdots \qquad \boxed{p = 2}$$

$$\sum_{k=1}^{\infty}\frac{1}{\sqrt{k}} = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{k}} + \cdots \qquad \boxed{p = \tfrac{1}{2}}$$

The following theorem tells when a p-series converges.

9.4.5 THEOREM (*Convergence of p-Series*)

$$\sum_{k=1}^{\infty}\frac{1}{k^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{k^p} + \cdots$$

converges if $p > 1$ and diverges if $0 < p \leq 1$.

PROOF To establish this result when $p \neq 1$, we will use the integral test.

$$\int_1^{+\infty}\frac{1}{x^p}\,dx = \lim_{b\to+\infty}\int_1^b x^{-p}\,dx = \lim_{b\to+\infty}\left.\frac{x^{1-p}}{1-p}\right]_1^b = \lim_{b\to+\infty}\left[\frac{b^{1-p}}{1-p} - \frac{1}{1-p}\right]$$

Assume first that $p > 1$. Then $1 - p < 0$, so $b^{1-p} \to 0$ as $b \to +\infty$. Thus, the integral converges [its value is $-1/(1-p)$] and consequently the series also converges.

Now assume that $0 < p < 1$. It follows that $1 - p > 0$ and $b^{1-p} \to +\infty$ as $b \to +\infty$, so the integral and the series diverge. The case $p = 1$ is the harmonic series, which was previously shown to diverge. ■

▶ Example 5

$$1 + \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{3}} + \cdots + \frac{1}{\sqrt[3]{k}} + \cdots$$

diverges since it is a p-series with $p = \frac{1}{3} < 1$. ◀

PROOF OF THE INTEGRAL TEST

Before we can prove the integral test, we need a basic result about convergence of series with *nonnegative* terms. If $u_1 + u_2 + u_3 + \cdots + u_k + \cdots$ is such a series, then its sequence of partial sums is increasing, that is,

$$s_1 \le s_2 \le s_3 \le \cdots \le s_n \le \cdots$$

Thus, from Theorem 9.2.3 the sequence of partial sums converges to a limit S if and only if it has some upper bound M, in which case $S \le M$. If no upper bound exists, then the sequence of partial sums diverges. Since convergence of the sequence of partial sums corresponds to convergence of the series, we have the following theorem.

9.4.6 THEOREM *If $\sum u_k$ is a series with nonnegative terms, and if there is a constant M such that*

$$s_n = u_1 + u_2 + \cdots + u_n \le M$$

for every n, then the series converges and the sum S satisfies $S \le M$. If no such M exists, then the series diverges.

In words, this theorem implies that *a series with nonnegative terms converges if and only if its sequence of partial sums is bounded above.*

PROOF OF THEOREM 9.4.4 We need only show that the series converges when the integral converges and that the series diverges when the integral diverges. For simplicity, we will limit the proof to the case where $a = 1$. Assume that $f(x)$ satisfies the hypotheses of the theorem for $x \ge 1$. Since

$$f(1) = u_1,\ f(2) = u_2, \ldots, f(n) = u_n, \ldots$$

the values of $u_1, u_2, \ldots, u_n, \ldots$ can be interpreted as the areas of the rectangles shown in Figure 9.4.2.

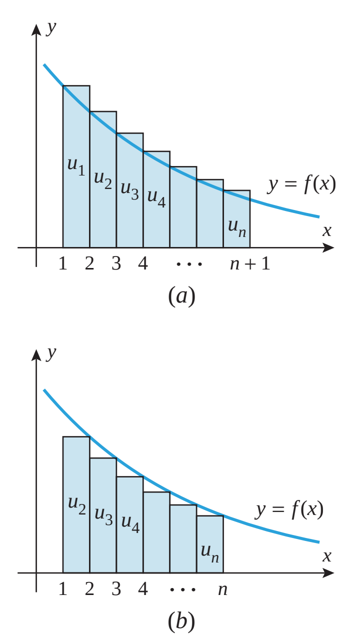


Figure 9.4.2

The following inequalities result by comparing the areas under the curve $y = f(x)$ to the areas of the rectangles in Figure 9.4.2 for $n > 1$:

$$\int_1^{n+1} f(x)\,dx < u_1 + u_2 + \cdots + u_n = s_n \qquad \text{Figure 9.4.2}a$$

$$s_n - u_1 = u_2 + u_3 + \cdots + u_n < \int_1^n f(x)\,dx \qquad \text{Figure 9.4.2}b$$

These inequalities can be combined as

$$\int_1^{n+1} f(x)\,dx < s_n < u_1 + \int_1^n f(x)\,dx \tag{2}$$

If the integral $\int_1^{+\infty} f(x)\,dx$ converges to a finite value L, then from the right-hand inequality in (2)

$$s_n < u_1 + \int_1^n f(x)\,dx < u_1 + \int_1^{+\infty} f(x)\,dx = u_1 + L$$

Thus, each partial sum is less than the finite constant $u_1 + L$, and the series converges by Theorem 9.4.6. On the other hand, if the integral $\int_1^{+\infty} f(x)\,dx$ diverges, then

$$\lim_{n\to+\infty} \int_1^{n+1} f(x)\,dx = +\infty$$

so that from the left-hand inequality in (2), $s_n \to +\infty$ as $n \to +\infty$. This implies that the series also diverges. ■

✔QUICK CHECK EXERCISES 9.4

(See page 631 for answers.)

1. The divergence test says that if ________ $\neq 0$, then the series $\sum u_k$ diverges.
2. Given that
$$a_1 = 3, \quad \sum_{k=1}^{\infty} a_k = 1, \quad \text{and} \quad \sum_{k=1}^{\infty} b_k = 5$$
it follows that
$$\sum_{k=2}^{\infty} a_k = \_\_\_\_\_\_\_\_ \quad \text{and} \quad \sum_{k=1}^{\infty}(2a_k + b_k) = \_\_\_\_\_\_\_\_$$
3. Since $\int_1^{+\infty}(1/\sqrt{x})\,dx = +\infty$, the ________ test applied to the series $\sum_{k=1}^{\infty}$ ________ shows that this series ________.
4. A p-series is a series of the form
$$\sum_{k=1}^{\infty} \_\_\_\_\_\_\_\_$$
This series converges if ________. This series diverges if ________.

EXERCISE SET 9.4 Graphing Utility C CAS

1. Use Theorem 9.4.3 to find the sum of each series.
(a) $\left(\frac{1}{2}+\frac{1}{4}\right)+\left(\frac{1}{2^2}+\frac{1}{4^2}\right)+\cdots+\left(\frac{1}{2^k}+\frac{1}{4^k}\right)+\cdots$
(b) $\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{5^k}-\frac{1}{k(k+1)}\right)$
2. Use Theorem 9.4.3 to find the sum of each series.
(a) $\displaystyle\sum_{k=2}^{\infty}\left[\frac{1}{k^2-1}-\frac{7}{10^{k-1}}\right]$ (b) $\displaystyle\sum_{k=1}^{\infty}\left[7^{-k}3^{k+1}-\frac{2^{k+1}}{5^k}\right]$

3–4 For each given p-series, identify p and determine whether the series converges. ■

3. (a) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k^3}$ (b) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt{k}}$ (c) $\displaystyle\sum_{k=1}^{\infty}k^{-1}$ (d) $\displaystyle\sum_{k=1}^{\infty}k^{-2/3}$
4. (a) $\displaystyle\sum_{k=1}^{\infty}k^{-4/3}$ (b) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt[4]{k}}$ (c) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt[3]{k^5}}$ (d) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k^{\pi}}$

5–6 Apply the divergence test and state what it tells you about the series. ■

5. (a) $\displaystyle\sum_{k=1}^{\infty}\frac{k^2+k+3}{2k^2+1}$ (b) $\displaystyle\sum_{k=1}^{\infty}\left(1+\frac{1}{k}\right)^k$
(c) $\displaystyle\sum_{k=1}^{\infty}\cos k\pi$ (d) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k!}$
6. (a) $\displaystyle\sum_{k=1}^{\infty}\frac{k}{e^k}$ (b) $\displaystyle\sum_{k=1}^{\infty}\ln k$
(c) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt{k}}$ (d) $\displaystyle\sum_{k=1}^{\infty}\frac{\sqrt{k}}{\sqrt{k}+3}$

7–8 Confirm that the integral test is applicable and use it to determine whether the series converges. ■

7. (a) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{5k+2}$ (b) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{1+9k^2}$
8. (a) $\displaystyle\sum_{k=1}^{\infty}\frac{k}{1+k^2}$ (b) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{(4+2k)^{3/2}}$

9–24 Determine whether the series converges. ■

9. $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k+6}$
10. $\displaystyle\sum_{k=1}^{\infty}\frac{3}{5k}$
11. $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt{k+5}}$
12. $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt[k]{e}}$
13. $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt[3]{2k-1}}$
14. $\displaystyle\sum_{k=3}^{\infty}\frac{\ln k}{k}$
15. $\displaystyle\sum_{k=1}^{\infty}\frac{k}{\ln(k+1)}$
16. $\displaystyle\sum_{k=1}^{\infty}ke^{-k^2}$
17. $\displaystyle\sum_{k=1}^{\infty}\left(1+\frac{1}{k}\right)^{-k}$
18. $\displaystyle\sum_{k=1}^{\infty}\frac{k^2+1}{k^2+3}$
19. $\displaystyle\sum_{k=1}^{\infty}\frac{\tan^{-1}k}{1+k^2}$
20. $\displaystyle\sum_{k=1}^{\infty}\frac{1}{\sqrt{k^2+1}}$
21. $\displaystyle\sum_{k=1}^{\infty}k^2\sin^2\left(\frac{1}{k}\right)$
22. $\displaystyle\sum_{k=1}^{\infty}k^2e^{-k^3}$
23. $\displaystyle\sum_{k=5}^{\infty}7k^{-1.01}$
24. $\displaystyle\sum_{k=1}^{\infty}\operatorname{sech}^2 k$

25–26 Use the integral test to investigate the relationship between the value of p and the convergence of the series. ■

25. $\displaystyle\sum_{k=2}^{\infty}\frac{1}{k(\ln k)^p}$
26. $\displaystyle\sum_{k=3}^{\infty}\frac{1}{k(\ln k)[\ln(\ln k)]^p}$

FOCUS ON CONCEPTS

27. Suppose that the series $\sum u_k$ converges and the series $\sum v_k$ diverges. Show that the series $\sum(u_k+v_k)$ and $\sum(u_k-v_k)$ both diverge. [*Hint:* Assume that $\sum(u_k+v_k)$ converges and use Theorem 9.4.3 to obtain a contradiction.]

28. Find examples to show that if the series $\sum u_k$ and $\sum v_k$ both diverge, then the series $\sum(u_k + v_k)$ and $\sum(u_k - v_k)$ may either converge or diverge.

29–30 Use the results of Exercises 27 and 28, if needed, to determine whether each series converges or diverges. ■

29. (a) $\displaystyle\sum_{k=1}^{\infty}\left[\left(\frac{2}{3}\right)^{k-1} + \frac{1}{k}\right]$ (b) $\displaystyle\sum_{k=1}^{\infty}\left[\frac{1}{3k+2} - \frac{1}{k^{3/2}}\right]$

30. (a) $\displaystyle\sum_{k=2}^{\infty}\left[\frac{1}{k(\ln k)^2} - \frac{1}{k^2}\right]$ (b) $\displaystyle\sum_{k=2}^{\infty}\left[ke^{-k^2} + \frac{1}{k\ln k}\right]$

31–34 True–False Determine whether the statement is true or false. Explain your answer. ■

31. If $\sum u_k$ converges to L, then $\sum(1/u_k)$ converges to $1/L$.

32. If $\sum cu_k$ diverges for some constant c, then $\sum u_k$ must diverge.

33. The integral test can be used to prove that a series diverges.

34. The series $\displaystyle\sum_{k=1}^{\infty}\frac{1}{p^k}$ is a p-series.

C **35.** Use a CAS to confirm that

$$\sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6} \quad\text{and}\quad \sum_{k=1}^{\infty}\frac{1}{k^4} = \frac{\pi^4}{90}$$

and then use these results in each part to find the sum of the series.

(a) $\displaystyle\sum_{k=1}^{\infty}\frac{3k^2-1}{k^4}$ (b) $\displaystyle\sum_{k=3}^{\infty}\frac{1}{k^2}$ (c) $\displaystyle\sum_{k=2}^{\infty}\frac{1}{(k-1)^4}$

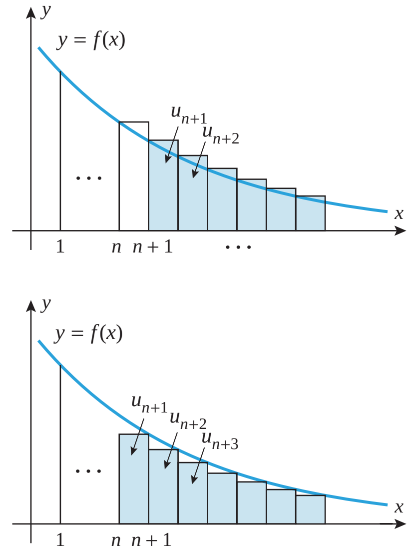
36–40 Exercise 36 will show how a partial sum can be used to obtain upper and lower bounds on the sum of a series when the hypotheses of the integral test are satisfied. This result will be needed in Exercises 37–40. ■

36. (a) Let $\sum_{k=1}^{\infty} u_k$ be a convergent series with positive terms, and let f be a function that is decreasing and continuous on $[n, +\infty)$ and such that $u_k = f(k)$ for $k \geq n$. Use an area argument and the accompanying figure to show that

$$\int_{n+1}^{+\infty} f(x)\,dx < \sum_{k=n+1}^{\infty} u_k < \int_{n}^{+\infty} f(x)\,dx$$

(b) Show that if S is the sum of the series $\sum_{k=1}^{\infty} u_k$ and s_n is the nth partial sum, then

$$s_n + \int_{n+1}^{+\infty} f(x)\,dx < S < s_n + \int_{n}^{+\infty} f(x)\,dx$$



◀ **Figure Ex-36**

37. (a) It was stated in Exercise 35 that

$$\sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}$$

Show that if s_n is the nth partial sum of this series, then

$$s_n + \frac{1}{n+1} < \frac{\pi^2}{6} < s_n + \frac{1}{n}$$

(b) Calculate s_3 exactly, and then use the result in part (a) to show that

$$\frac{29}{18} < \frac{\pi^2}{6} < \frac{61}{36}$$

(c) Use a calculating utility to confirm that the inequalities in part (b) are correct.

(d) Find upper and lower bounds on the error that results if the sum of the series is approximated by the 10th partial sum.

38. In each part, find upper and lower bounds on the error that results if the sum of the series is approximated by the 10th partial sum.

(a) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{(2k+1)^2}$ (b) $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k^2+1}$ (c) $\displaystyle\sum_{k=1}^{\infty}\frac{k}{e^k}$

39. It was stated in Exercise 35 that

$$\sum_{k=1}^{\infty}\frac{1}{k^4} = \frac{\pi^4}{90}$$

(a) Let s_n be the nth partial sum of the series above. Show that

$$s_n + \frac{1}{3(n+1)^3} < \frac{\pi^4}{90} < s_n + \frac{1}{3n^3}$$

(b) We can use a partial sum of the series to approximate $\pi^4/90$ to three decimal-place accuracy by capturing the